# Sales Analytics Case Study

## Dataset

```
sales.txt

1001 electronics 1200
1002 clothing 800
1003 electronics 500
1004 groceries 300
1005 clothing 400
1006 electronics 900
```

Columns

```
OrderID   Category   Amount
```

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# Task 1 — Total Revenue per Category

### Objective

Calculate the **total revenue generated by each product category**.

### Mapper Output

```
(category, amount)
```

### Reducer Logic

```
sum(amount)
```

### Expected Output

```
electronics 2600
clothing 1200
groceries 300
```

---

# Task 2 — Number of Orders per Category

### Business Objective

The company wants to know **how many orders were placed for each product category**.

### Mapper Output

```
(category, 1)
```

### Reducer Logic

```
count orders
```

### Expected Output

```
electronics 3
clothing 2
groceries 1
```

---

# Task 3 — Average Order Value per Category

### Business Objective

The analytics team wants to calculate the **average order value for each product category**.

### Mapper Output

```
(category, amount)
```

### Reducer Logic

Reducer must compute:

```
average = total_amount / number_of_orders
```

### Expected Output

```
electronics 866.67
clothing 600
groceries 300
```

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# Task 4 — Identify the Highest Revenue Category

### Business Objective

Management wants to identify **which category generates the highest total revenue**.

### Steps

1. Compute revenue per category
2. Sort output

3. Return the top category

### Expected Output

```
electronics 2600
```

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# Task 5 — Filter High-Value Transactions

### Business Objective

The company wants to analyze **transactions where order value is greater than 700**.

### Mapper Logic

Filter records where

```
amount > 700
```

### Expected Output

```
1001 electronics 1200
1002 clothing 800
1006 electronics 900
```

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# Task 6 — Category Contribution Percentage

### Business Objective

Calculate the **percentage contribution of each category to total revenue**.

### Step 1

Calculate total revenue of all orders.

```
Total Revenue = 4100
```

### Step 2

Compute contribution

```
electronics = 63.41%
clothing = 29.26%
groceries = 7.31%
```

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# Task 7 — Identify High Revenue Orders

### Business Objective

Find **orders whose value is above the category average**.

Example

```
electronics avg = 866
```

High-value order

```
1001 electronics 1200
1006 electronics 900
```

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# Task 8 — Top 2 Orders per Category

### Business Objective

Identify the **top 2 highest value orders in each category**.

### Expected Output Example

```
electronics 1200
electronics 900
clothing 800
clothing 400
groceries 300
```

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